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General Characteristics of Inner-transition Elements : *f*-Block Elements (Lanthanides and Actinides)

Introduction

The elements from Ce_{58} to La_{71} (14 elements) and from Th_{90} to Lw_{103} (14 elements) are called **inner-transition elements.** These elements have been shown at a separate place at the bottom of the periodic table. The first series of 14 elements lies between La_{57} and Hf_{72} in 6th period while the other series of 14 elements lies between Ac_{89} and Ku_{104} in 7th period (incomplete period). Both the series are present in IIIB (3) group.

The elements of 1st series are called **lanthanides** or **lanthanones** (rare earth elements) while those of the 2nd series are called **actinides** or **actinones.** The names lanthanides and actinides have been given, since these elements follow La_{57} and Ac_{89} respectively with which these elements show close similarities.

Since the last electron (differentiating electron) in the atoms of these elements enters the *f*-sub-shell belonging to anti-penultimate shell [*i.e.*, $(n - 2)$th shell], these elements are also called *f*-**block elements.** Thus the last electron enters $(n - 2)f$ sub-shell in inner-transition elements.

Since lanthanides are present in 6th period ($n = 6$) of the periodic table, the last electron in these elements goes to $(6 - 2)f$ or $4f$ sub-shell. Thus lanthanides are also called $4f$ **block elements,** *i.e.*, *lanthanides can be defined as those elements in which the last electron goes to 4f sub-shell.*

Similarly, since actinides are present in 7th period ($n = 7$) of the periodic table, the last electron in these elements goes to $(7 - 2)f$ or $5f$ sub-shell. Thus actinides are also called **5*f*-block elements**, *i.e.*, *actinides are those elements in which the last electron enters 5f sub-shell.*

Valence-shell electronic configuration of the atoms of *f*-block elements can be represented as $(n - 2)f^{0-14}$, $(n - 1)d^{0-2}$, ns^2 which shows that in these elements the outermost three shells are partially-filled while the remaining inner shells are completely-filled.

Now we shall discuss lanthanides ($4f$-block elements) and actinides ($5f$-block elements) under separate sections.

Lanthanides: 4*f*-Block Elements

As already said, the 14 elements *viz.* Ce_{58} to Lu_{71} are called lanthanides.

General characteristics of lanthanides

The following are important properteis of lanthanides.

observed and expected configuration of lanthanides ?

1. Electronic configurations

We know that the electronic configuration of La_{57} which is followed by 14 lanthanides is $[Xe]_{54}\ 4f^0\ 5d^1\ 6s^2$ in which $5d$ sub-shell is singly-filled and $4f$ sub-shell is vacant. This configuration shows that as we move in the series of 14 lanthanides (Ce_{58} to Lu_{71}), the additional electron should occupy the vacant $4f$ orbitals and $5d$ orbitals should remain singly-filled, *i.e.*, the expected configurations of the atoms of lanthanides should be those in which $5d$ orbitals are singly-filled and $4f$ orbitals are progressively filled up with electrons. In other words, we can say that the expected configuration of the atoms of lanthanides should be $[Xe]_{54}\ 4f^{1\text{-}14}\ 5d^1\ 6s^2$. But, since the energies of $5d$ and $4f$ orbitals are closely similar, in all the lanthanides, excepting Gd_{64} and Lu_{71}, $5d^1$ electron gets shifted to 4f orbitals and hence $5d$ orbitals remain vacant. In Gd_{64}, the shifting of $5d^1$ electron to $4f$ orbitals does not take place, since this type of shifting gives unstable configuration *viz.* $[Xe]_{54}\ 5f^8 5d^0 6s^2$ to Gd_{64}. In Lu_{71} the shifting of $5d^1$ electron to $4f$ orbitals is also not possible because $4f$ orbitals are already filled to their maximum capacity of 14 electrons.

Observed as well as expected configurations are given in Table 10.1. Observed configurations show that the complete and valence configurations can be written as :

Complete configuration : 2, 8, 18, 20 to 32, 8 or 9, 2

or 2, 8, 18, $4s^2p^6d^{10}f^{2\text{-}14}$, $5s^2p^6d^{0\text{-}1}$, $6s^2$

or $[Xe]_{54}\ 4f^{2\text{-}14}\ 5d^{0\text{-}1}6s^2$

Valence-shell configuration : $4f^{2\text{-}14}\ 5d^{0\text{-}1}\ 6s^2$.

2. Oxidation states

Main oxidation states exhibited by lanthanides are shown in Table 10.2. This table shows that +3 is the principal oxidation state which is shown by all the lanthanides. Although a few lanthanides also show +2 and +4 oxidation states, +3 is the most stable oxidation state for all the lanthanides since some of the M^{2+} and M^{4+} cations are converted into M^{3+} ions, *e.g.*, Sm^{2+} is a good reducing agent while Ce^{4+} is a good oxidising agent, since both these ions are converted into M^{3+} ions which are the most stable ions. This is evident from the following reactions.

$$\underset{(H = +1)}{2Sm^{2+}} + 2H_2O \longrightarrow 2Sm^{3+} + 2OH^- + \underset{(H = 0)}{H_2}$$

$$Ce^{4+} + Fe^{2+} \longrightarrow Ce^{3+} + Fe^{3+}$$

For some lanthanides, +2, +3 and +4 oxidation states are explained on the basis of the fact that M^{2+}, M^{3+} and M^{4+} ions attain $4f^0$, $4f^7$ and $4f^{14}$ configurations respectively which are very stable configurations. For example:

(*i*) La and Ce attain $4f^0$ configuration when they are in +3 and +4 oxidation states respectively.

$$La^{3+} = [Xe]_{54}\ 4f^0; \qquad Ce^{4+} = [Xe]_{54}\ 4f^0$$

(*ii*) Eu, Gd and Tb get $4f^7$ configuration in +2, +3 and +4 oxidation states respectively.

$$Eu^{2+} = [Xe]_{54}\ 4f^7; \qquad Gd^{3+} = [Xe]_{54}\ 4f^7; \qquad Tb^{4+} = [Xe]_{54}\ 4f^7$$

(*iii*) Yb and Lu show +2 and +3 oxidation states, since these oxidation states have $4f^{14}$ configuration

$$Yb^{2+} = [Xe]_{54}\ 4f^{14}; \qquad Lu^{3+} = [Xe]_{54}\ 4f^{14}$$

Exceptions. Many of the lanthanides show +2, +3 and +4 oxidation states but the elements in these oxidation states *donot* have $4f^0$, $4f^7$ and $4f^{14}$ configurations. For example :

Ce^{2+}	$= [Xe]_{54}\ 4f^2$	Sm^{2+}	$= [Xe]_{54}\ 4f^6$
Tm^{2+}	$= [Xe]_{54}\ 4f^{13}$	Ce^{3+}	$= [Xe]_{54}\ 4f^1$
Pr^{3+}	$= [Xe]_{54}\ 4f^2$	Pr^{4+}	$= [Xe]_{54}\ 4f^1$
Nd^{4+}	$= [Xe]_{54}\ 4f^2$	Dy^{2+}	$= [Xe]_{54}\ 4f^8$

The above oxidation states can be explained on the basis of thermodynamic and kinetic factors.

Table 10.1. *Expected and observed (actual) electronic configurations of lanthanide atoms. The configuration of La_{57} is also given. $[Xe]_{54}$ = 2, 8, 18, 18, 8*

Lanthanide element	*Expected configuration*	*Observed (actual) configuration*
Lanthanum, La_{57}	$[Xe]_{54}\ 4f^1 5d^0 6s^2$ = 2, 8, 18, (18 + 1), 8, 2 = 2, 8, 18, 19, 8, 2	$[Xe]_{54}\ 4f^0\ 5d^1\ 6s^2$ = 2, 8, 18, (18 + 0), (8 + 1), 2 = 2, 8, 18, 18, 9, 2
Cerium, Ce_{58}	$[Xe]_{54}\ 4f^1 5d^1 6s^2$ = 2, 8, 18, (18 + 1), (8 + 1), 2 = 2, 8, 18, 19, 8, 2	$[Xe]_{54}\ 4f^2 5d^0 6s^2$ = 2, 8, 18, (18 + 2), 8, 2 = 2, 8, 18, 20, 8, 2
Praseodymium, Pr_{59}	$[Xe]_{54}\ 4f^2 5d^1 6s^2$ = 2, 8, 18, (18 + 2), 8, 2 = 2, 8, 18, 20, 9, 2	$[Xe]_{54}\ 4f^3 5d^0 6s^2$ = 2, 8, 18, (18 + 3), 8, 2 = 2, 8, 18, 21, 8, 2
Neodymium, Nd_{60}	$[Xe]_{54}\ 4f^3 5d^1 6s^2$ = 2, 8, 18, (18 + 3), 9, 2 = 2, 8, 18, 21, 9, 2	$[Xe]_{54}\ 4f^4\ 5d^0\ 6s^2$ = 2, 8, 18, (18 + 4), 8, 2 = 2, 8, 18, 22, 8, 2
Promethium, Pm_{61}	$[Xe]_{54}\ 4f^4 5d^1 6s^2$ = 2, 8, 18, (18 + 4), 9, 2 = 2, 8, 18, 22, 9, 2	$[Xe]_{54}\ 4f^5\ 5d^0\ 6s^2$ = 2, 8, 18, (18 + 5), 8, 2 = 2, 8, 18, 23, 8, 2
Samarium, Sm_{62}	$[Xe]_{54}\ 4f^5 5d^1 6s^2$ = 2, 8, 18, (18 + 5), 9, 2 = 2, 8, 18, 23, 9, 2	$[Xe]_{54}\ 4f^6\ 5d^0\ 6s^2$ = 2, 8, 18, (18 + 6), 8, 2 = 2, 8, 18, 24, 8, 2
Europium, Eu_{63}	$[Xe]_{54}\ 4f^6 5d^1 6s^2$ = 2, 8, 18, (18 + 6), 9, 2 = 2, 8, 18, 24, 9, 2	$[Xe]_{54}\ 4f^7\ 5d^0\ 6s^2$ = 2, 8, 18, (18 + 7), 8, 2 = 2, 8, 18, 25, 8, 2
Gadolinium, Gd_{64}	$[Xe]_{54}\ 4f^7 5d^1 6s^2$ = 2, 8, 18, (18 + 7), 9, 2 = 2, 8, 18, 25, 9, 2	$[Xe]_{54}\ 4f^7\ 5d^1\ 6s^2$ = 2, 8, 18, (18 + 7), 9, 2 = 2, 8, 18, 25, 8, 2
Terbium, Tb_{65}	$[Xe]_{54}\ 4f^8 5d^1 6s^2$ = 2, 8, 18, (18 + 8), 9, 2 = 2, 8, 18, 26, 9, 2	$[Xe]_{54}\ 4f^9\ 5d^0\ 6s^2$ = 2, 8, 18, (18 + 9), 8, 2 = 2, 8, 18, 27, 8, 2
Dysprosium, Dy_{66}	$[Xe]_{54}\ 4f^9 5d^1 6s^2$ = 2, 8, 18, (18 + 9), 9, 2 = 2, 8, 18, 27, 9, 2	$[Xe]_{54}\ 4f^{10}\ 5d^0\ 6s^2$ = 2, 8, 18, (18 + 10), 8, 2 = 2, 8, 18, 28, 8, 2
Holmium, Ho_{67}	$[Xe]_{54}\ 4f^{10} 5d^1 6s^2$ = 2, 8, 18, (18 + 10), 9, 2 = 2, 8, 18, 28, 9, 2	$[Xe]_{54}\ 4f^{11}\ 5d^0\ 6s^2$ = 2, 8, 18, (18 + 11), 8, 2 = 2, 8, 18, 29, 8, 2
Erbium, Er_{68}	$[Xe]_{54}\ 4f^{11} 5d^1 6s^2$ = 2, 8, 18, (18 + 11), 9, 2 = 2, 8, 18, 29, 9, 2	$[Xe]_{54}\ 4f^{12}\ 5d^0\ 6s^2$ = 2, 8, 18, (18 + 12), 8, 2 = 2, 8, 18, 30, 8, 2
Thulium, Tm_{69}	$[Xe]_{54}\ 4f^{12} 5d^1 6s^2$ = 2, 8, 18, (18 + 12), 9, 2 = 2, 8, 18, 30, 9, 2	$[Xe]_{54}\ 4f^{13}\ 5d^0\ 6s^2$ = 2, 8, 18, (18 + 13), 8, 2 = 2, 8, 18, 31, 8, 2
Ytterbium, Yb_{70}	$[Xe]_{54}\ 4f^{13} 5d^1 6s^2$ = 2, 8, 18, (18 + 13), 9, 2 = 2, 8, 18, 31, 9, 2	$[Xe]_{54}\ 4f^{14}\ 5d^0\ 6s^2$ = 2, 8, 18, (18 + 14), 8, 2 = 2, 8, 18, 32, 8, 2
Lutetium, Lu_{71}	$[Xe]_{54}\ 4f^{14} 5d^1 6s^2$ = 2, 8, 18, (18 + 14), 9, 2 = 2, 8, 18, 32, 9, 2	$[Xe]_{54}\ 4f^{14}\ 5d^1\ 6s^2$ = 2, 8, 18, (18 + 14), 9, 2 = 2, 8, 18, 32, 9, 2

3. Atomic and ionic radii of M^{3+} ions: Lanthanide contraction

Atomic and ionic radii of M^{3+} ions have already been given in Table 10.2. A look at these values shows that there in a *steady* decrease in these values as we move from Ce to Lu and from Ce^{3+} to Lu^{3+}.

Table 10.2. *Oxidation states, atomic radii and ionic radii (of M^{3+} ions) of lanthanides.*

Lanthanides	*Oxidation states*	*Atomic radii (pm)*	*Ionic radii (M^{3+} ions) (pm)*
La	+3	169	106
√ Ce	+3, +4	165	103
√ Pr	+3, +4	165	101
Nd	+2, +3, +4	164	100
Pm	+3	—	98
● Sm	+2, +3	166	96
● Eu	+2, +3	185	95
Gd	+3	161	94
√ Tb	+3, +4	159	92
√ Dy	+3, +4	159	91
Ho	+3	158	89
Er	+3	157	88
● Tm	+2, +3	156	87
● Yb	+2, +3	170	86
Lu	+3	156	85

The steady decrease in the atomic and ionic radii (M^{3+} ions) of lanthanide elements with increasing atomic number is called **lanthanide contraction.**

When we say that there is a steady decrease in the values of atomic and ionic radii, this means that the decrease is *very small*. For example on moving from Ce to Lu the atomic radii decrease from 165 pm to 156 pm and the decrease is only equal to 165 – 156 = 9 pm. Similarly, the ionic radii decrease from 103 to 85 on moving from Ce^{3+} to Lu^{3+} and the decrease in only equal to 103 – 85 = 18 pm. Thus we see that although the atomic number or nuclear charge increases by +14 units which is a large number, the decrease in atomic and ionic radii is only equal to 9 pm and 18 pm respectively. This decrease in atomic and ionic radii is very small as compared to that found in the elements of other groups and periods.

It may be seen from the table that although the atomic radii show some irregularities (*e.g.*, the atomic radii of Eu and Yb are abnormally high), the ionic radii of M^{3+} ions contract steadily.

Cause of lanthanide contraction

We know that as we proceed from one element to the next one in lanthamide series, the nuclear charge (*i.e.*, atomic number) increases by +1 at each next element. Thus as we move from Ce to Lu, the attraction between the nucleus and the outermost shell electron increases gradually at each step. It is also known that as we move from Ce to Lu, the addition of extra electron takes place to 4*f* orbitals. Since 4*f* orbitals have very diffused shape, the electrons in these orbitals are not able to shield (decrease) effectively the attraction of the nucleus for the

It may be seen from the table that the colour depends on the number of electrons present in 4*f* orbitals. The ion having *n* electrons in 4*f* orbitals has the same colour as the ion which has (14 – *n*) electrons in 4*f* orbitals. For example La^{3+} ($4f^0$) which has no electron in its 4*f* orbitals ($n = 0$) is colourless and Lu^{3+} ion ($4f^{14}$) which has (14 – 0) = 14 electrons in 4*f* orbitals is also colourless. Similarly Pr^{3+} ion ($4f^2$) which has two electrons in its 4*f* orbitals ($n = 2$) and Tm^{3+} ion ($4f^{12}$) which has (14 – 2) = 12 electrons in its 4*f* orbitals have the same colour (green). (See Fig. 10.1)

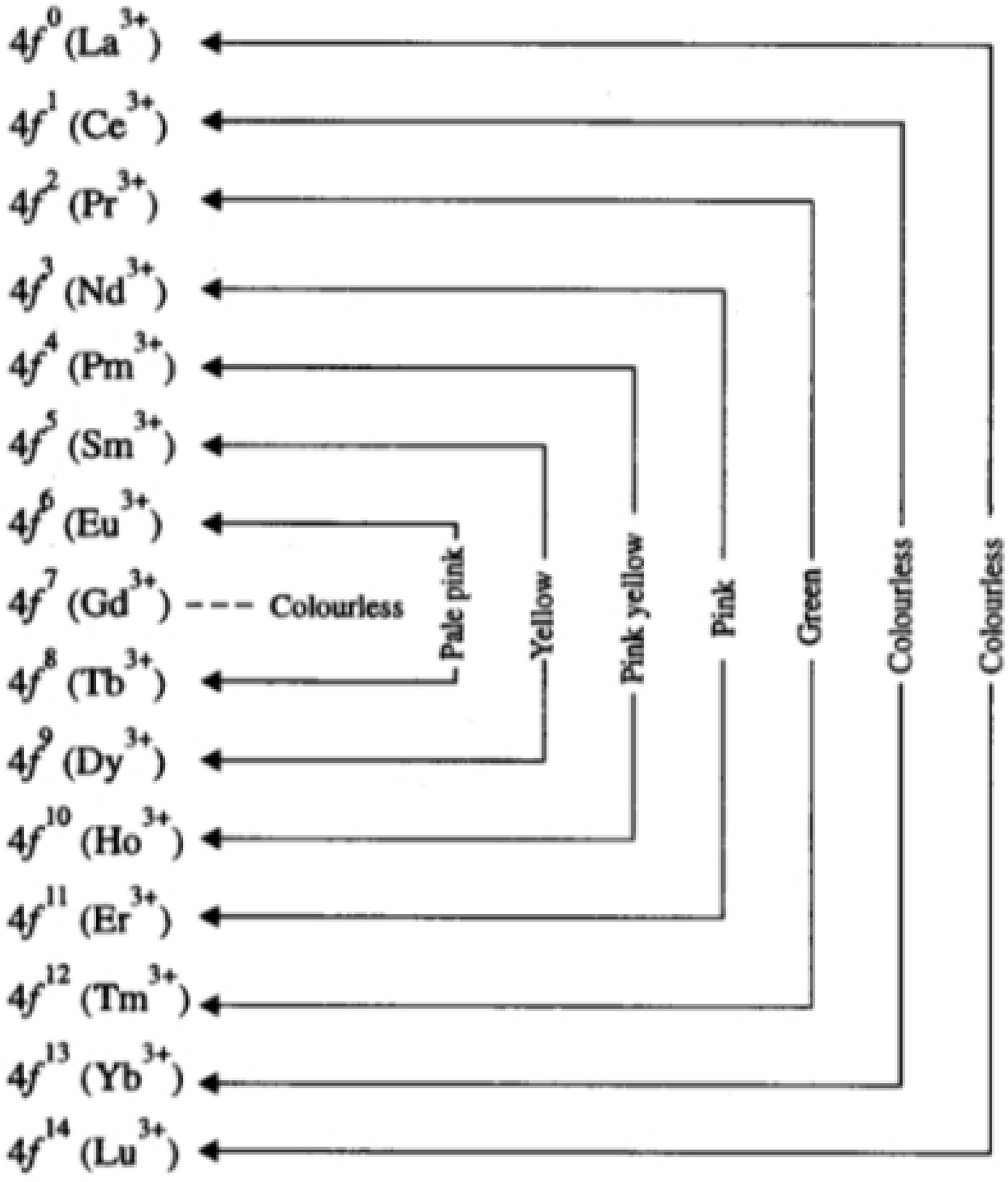


Fig. 10.1. Colour of tripositive lanthanide cations (M^{3+} ions).

5. Magnetic properties of M^{3+} ions

We know that ions having unpaired electrons are paramagnetic while those having all the electrons paired are diamagnetic. Thus, since La^{3+} ($4f^0.5d^0.6s^0$) and Lu^{3+} ($4f^{14}\ 5d^0\ 6s^0$) ions have all the electrons paired, these ions are diamagnetic. On the other hand, the remaining tripositive lanthanide ions (M^{3+} ions) are paramagnetic. since they contain 1, 2, ...7 unpaired electrons.

Cause of paramagnetism

In case of M^{3+} cations, since 4*f* orbitals are quite deeply seated, these are well shielded from the ligands surrounding the M^{3+} ions by the overlapping 5*s* and 5*p* orbitals and hence the electrons residing in 4*f* orbitals are not able to interact with the surrounding ligands. Thus the orbital motion of electrons in 4*f* orbitals is not quenched as in the case of *d*-block elements, and hence the observed paramagnetism in 4*f*-block elements is due to both factors namely orbital motion of the electron and its spinning round its axis.

Calculated an experimental values of magnetic moment (in BM) of the sulphates of triposotive lanthanide ions (M^{3+} ions) are given in Table 10.4 In this table *n* indicates the number of unpaired electrons and J is the total angular momentum. This table shows that the calculated value increases

5:55 79%

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upto Nd^{3+} (= 3.62) and then drops to 0.84 for Sm^{3+}. It starts rising again, becoming maximum for Dy^{3+} (= 10.65). It again starts decreasing becoming zero for Lu^{3+} which is diamagnetic.

Table 10.4. *Calculated and experimental values of magnetic moment (in BM) of the sulphates of M^{3+} ions (n = No. of unpaired electrons in 4f-orbitals)*

M^{3+} ion	*$4f^x$ electrons*	*n*	*Calculated value, $\mu_{cal} = g\sqrt{J(J+1)}$ BM (Theoretical values)*	*Experimental value, μ_{exp} in (BM)*
La^{3+}	$4f^0$ ($x = 0$)	0	0.00	Diamagnetic
Ce^{3+}	$4f^1$	1	2.54	2.3–2.5
Pr^{3+}	$4f^2$	2	3.58	3.4–3.6
Nd^{3+}	$4f^3$	3	3.62	3.5–3.6
Pm^{3+}	$4f^4$	4	2.70	—
Sm^{3+}	$4f^5$	5	0.84	1.5–1.6
Eu^{3+}	$4f^6$	6	0.00	3.4–3.6
Gd^{3+}	$4f^7$	7	7.94	7.8–8.0
Tb^{3+}	$4f^8$	6	9.70	9.4–9.6
Dy^{3+}	$4f^9$	5	10.65	10.4–10.5
Ho^{3+}	$4f^{10}$	4	10.60	10.3–10.5
Er^{3+}	$4f^{11}$	3	9.60	9.4–9.6
Tm^{3+}	$4f^{12}$	2	7.60	7.1–7.4
Yb^{3+}	$4f^{13}$	1	4.50	4.4–4.9
Lu^{3+}	$4f^{14}$ ($x = 14$)	0	0.00	Diamagnetic

6. Physical properties

All the lanthanides are soft, malleable and ductile, and have low tensile strength. They are not good conductors of heat and electricity. In general, the atomic volumes and densities of these elements increase with the increase in atomic number.

7. Chemical reactivity of lanthanides

All the lanthanides are almost equally chemical reactive. Their similar chemical reactivity is due to the fact that since 4*f* electrons in lanthanides are very effectively shielded from the interaction with other elements by the overlapping 5*s*, 5*p* and 6*s* electrons (5*d* orbitals donot contain any electron), these elements have very little difference in their chemical reactivity. It is because of similarity in their chemical reactivity that lanthanides occur together in nature and hence it is difficult to separate these elements from each other.

Lanthanides are generally more reactive them *d*-block elements.

8. Formation of compounds

(*a*) Many lanthanides react with carbon to form salt-like carbides and with H_2 to give salt- like hydrides. Lanthanides react with air or O_2 and S to form sesquioxides (M_2O_3) and sulphides (M_2S_3) respectively. Cerium gives CeO_2. Ytterbium, however, resists the action of air even at 1000°C due to the formation of a protective coating of its oxide. M_2O_3 oxides react with H_2O to form in soluble hydroxides, $M(OH)_3$. Oxides react with CO_2 to give carbonates, $M_2(CO_3)_3$.

All the lanthanides react with H_2O slowly in cold but rapidly on heating and liberate H_2 and form $M(OH)_3$.

(*b*) Lanthanide compounds are generally ionic, coloured and show paramagnetism. Fluorides, oxides, hydroxides, carbonates, phosphates, chromates and oxalates of lanthanides are largely soluble in water. On the other hand, the halides other than fluorides, nitrates, acetates, perchlorates and salts of oxy acids of lanthanides follow the pattern of solubility of the salts of group 2 elements. However, lanthanide sulphates are soluble in water.

Q⇒ All hydroxides, $M(OH)_3$ are ionic and basic in nature. These are stronger bases than $Al(OH)_3$, but weaker than $Ca(OH)_2$. Due to lanthanide contraction, the basicity of the hydroxides decreases from $La(OH)_3$ to $Lu(OH)_3$. Thus $La(OH)_3$ is the strongest base and $Lu(OH)_3$ is the weakest base.

Hydrides react with CO_2 to form $M_2(CO_3)_3$.

9. Formation of complexes

Although tripositive lanthanide cations have a high charge equal to +3 on them, yet their size is so large that their charge-to-radius ratio becomes so small that these ions have very poor tendency to form complexes. Common ligands with which M^{3+} ions form stable complexes are: (*i*) *chelating oxygen containing ligands* like EDTA citric acid, oxalic acid, acetylacetone, (*ii*) *nitrogen containing ligands* like ethylenediamine, NCS etc.

10. Properties dependent on standard reduction potential values

The values of standard reduction potential (E^0 values) for the reduction reaction:

$$M^{3+}(aq) + 3e^- \rightleftharpoons M(s)$$

increase from La to Lu, *i.e.*, E^0 values becomes less negative as we move from La to Lu as shown below (values are given in volts):

$\longleftarrow$ *E^o values become less negative* $\longrightarrow$

La = – 2.52, Ce = –2.48, Pr = – 2.46, Nd = – 2.43, Pm = – 2.42, Sm = – 2.41, Eu = – 2.40, Gd = –2.39, Tb = – 2.39, Dy = – 2.35, Ho = – 2.32, Er = – 2.30, Tm = – 2.28, Yb = – 2.27, Lu = – 2.25.

These E^o values explain the following properties of lanthanides:

(*a*) **Reducing property.** Since lanthanides have high negative E^o values, these elements have a strong tendency to lose their three electrons to undergo oxidation and hence are strong reducing agents. With the increase of E^0 values from La to Lu, *i.e.*, as E^o values become less negative from La to Lu, the reducing power of these elements decreases from La to Lu.

(*b*) **Electropositive character.** Due to negative E^0 values, the lanthanides readily lose their electrons and hence show strong electropositive (or metallic) character. With the increase of E^o values from La to Lu, the electropositive character of these elements decreases from La to Lu.

(*c*) **Liberation of H_2 from H_2O.** Due to highly electropositive character of lanthanides, these elements slowly react with cold water and rapidly with hot water to liberate H_2 from it.

Extraction of lanthanides from monazite sand

The method used for the extraction of lanthanides from monazite sand consists of the steps which have been shown in flow sheet 1 given on page 346. The solution obtained after removing cerium as basic ceric nitrate, $Ce(OH)(NO_3)_3.3H_2O$ contains the nitrates of La, Pr, Nd and Sm. The individual lanthanides are separated from this solution by suitable methods.

Separation of lanthanide elements

Q = Methods for the separation of lanthanides.

All the lanthanide elements form tripositive ions (M^{3+} ions) which are almost identical in size. The chemical properties of lanthanides which depend on the size and charge of M^{3+} ions are almost the same. Due to the similarity in chemical properties, it is difficult to separate the lanthanides from each other. However, the following methods have been used to separate them.